



Scholarly Migration Database (SMD)

and example projects using it

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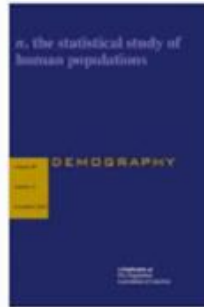
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A SIMPLE IDEA

Changes in institutional affiliations can be used to infer changes in residence over time for individual scholars and for populations



Demography.

November 2012, Volume 49, [Issue 4](#), pp 1307–1333 | [Cite as](#)

Migration Systems in Europe: Evidence From Harmonized Flow Data

[Authors](#)

[Authors and affiliations](#)

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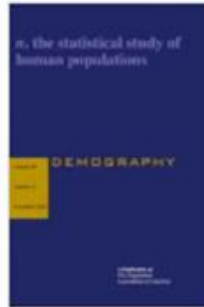
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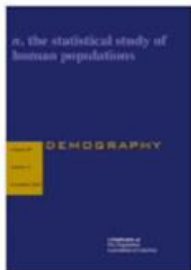
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Demography.

June 2015, Volume 52, Issue 3, pp 1035–1059 | [Cite as](#)

Bayesian Population Forecasting: Extending the Lee-Carter Method

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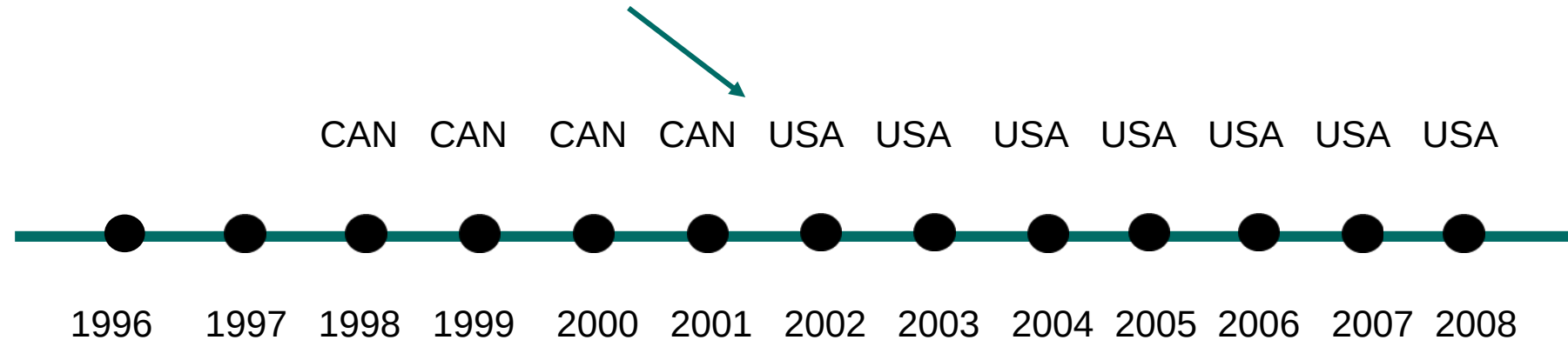
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IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLE

Maria changes residence
from Canada to the US
between 2001 and 2002



Maria's **modal countries**
of publications over time

A SCALABLE IDEA: THE SCOPUS DATABASE



90M+
records
(to 1788)



1.8B+
cited references
(to 1970)



17M+
author profiles



292K+
stand-alone
books



7K+
publishers



94K+
affiliation profiles

Data accessed via:



MAX PLANCK
digital library



OpenAlex

Source: <https://www.elsevier.com/products/scopus/content> and <https://blog.scopus.com/posts/the-scopus-content-coverage-guide-a-complete-overview-of-the-content-coverage-in-scopus-and>

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SMD paper and data



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scientific data

OPEN

DATA DESCRIPTOR

Bilateral flows and rates of international migration of scholars for 210 countries for the period 1998-2020

Aliakbar Akbaritabar^{1,2}, Tom Theile^{1,2} & Emilio Zagheni¹

A lack of comprehensive migration data is a major barrier for understanding the causes and consequences of migration processes, including for specific groups like high-skilled migrants. We leverage large-scale bibliometric data from Scopus and OpenAlex to trace the global movements of scholars. Based on our empirical validations, we develop pre-processing steps and offer best practices for the measurement and identification of migration events. We have prepared a publicly accessible dataset that shows a high level of correlation between the counts of scholars in Scopus and OpenAlex for most countries. Although OpenAlex has more extensive coverage of non-Western countries, the highest correlations with Scopus are observed in Western countries. We share aggregated yearly estimates of international migration rates and of bilateral flows for 210 countries and areas worldwide for the period 1998–2020 and describe the data structure and usage notes. We expect that the publicly shared dataset will enable researchers to further study the causes and the consequences of migration of scholars to forecast the future mobility of academic talent worldwide.

Paper: <https://doi.org/10.1038/s41597-024-03655-9>

Data: <https://doi.org/10.5281/zenodo.11145735>

zenodo

Search records... Communities My dashboard

Published May 8, 2024 | Version v1

Dataset Open

Replication data for: Bilateral flows and rates of international migration of scholars for 210 countries and areas for the period 1998-2020

Theile, Tom (Other)¹; Akbaritabar, Aliakbar¹; Zagheni, Emilio¹

Show affiliations

Contributors

Data curators: Akbaritabar, Aliakbar¹; Theile, Tom¹

Data manager: Zagheni, Emilio¹

Show affiliations

Data and code for performing analyses and plotting figures for "Bilateral flows and rates of international migration of scholars for 210 countries and areas for the period 1998-2020"

The code and data can also be found at <https://github.com/MPIDR/Global-flows-and-rates-of-international-migration-of-scholars/>

Abstract A lack of comprehensive migration data is a major barrier for understanding the causes and consequences of migration processes, including for specific groups like high-skilled migrants. We leverage large-scale bibliometric data from Scopus and OpenAlex to trace the global movements of scholars. Based on our empirical validations, we develop pre-processing steps and offer best practices for the measurement and identification of migration events. We have prepared a publicly accessible dataset that shows a high level of correlation between the counts of scholars in Scopus and OpenAlex for most countries. Although OpenAlex has more extensive coverage of non-Western countries, the highest correlations with Scopus are observed in Western countries. We share aggregated yearly estimates of international migration rates and of bilateral flows for 210 countries and areas worldwide for the period 1998-2020 and describe the data structure and usage notes. We expect that the publicly shared dataset will enable researchers to further study the causes and the consequences of migration of scholars to forecast the future mobility of global academic talent.

Files

Global-flows-and-rates-of-international-migration-of-scholars-2024_v1.1.zip

Global-flows-and-rates-of-international-migration-of-scholars-2024_v1.1.zip

- Global-flows-and-rates-of-international-migration-of-scholars-2024_v1.1
 - .gitignore 448 Bytes
 - 01_prepare_enrich_data.py 10.6 kB
 - 02_merge_openalex_and_scopus.py 6.2 kB
 - 03_plotting.py 17.4 kB

Scholarly Migration Database

- Visit:
 - <https://www.scholarlymigration.org/>
- To contact SMD's team
 - Scholarlymigration@demogr.mpg.de
 - <https://twitter.com/SMDatabase>



Scholarly Migration Database

Quantifying the Mobility of Scholars



The Future

- Include Gender
- Include subnational regions
- Make more fine grained data available
- Use and combine with OpenAlex and ORCID



Some of the projects using SMD

A gender perspective on the global migration of scholars

Xinyi Zhao ^{a,b,1}, Aliakbar Akbaritabar ^a, Ridhi Kashyap ^{b,c}, and Emilio Zagheni^a

Edited by Douglas Massey, Princeton University, Princeton, NJ; received August 26, 2022; accepted January 5, 2023

February 27, 2023 | 120 (10) e2214664120 | <https://doi.org/10.1073/pnas.2214664120>

Significance

Within a globalizing scientific system, international migration is increasingly recognized as a strategy for scientists to advance their careers. The migration literature more broadly has suggested a process of feminization, with an increasing share of women among all international migrants. With respect to the migration of scholars, however, whether male and female scholars participate equally in transnational mobility and how these patterns have shifted over time from a global perspective are not known. Using bibliometric data that cover the past two decades, we show that, while female researchers continued to be underrepresented among internationally mobile researchers, and migrated over shorter distances, this gender gap has been narrowing at a faster rate than the gender gap in the population of general researchers.

- Female researchers continued to be **underrepresented in transnational scholarly mobility**, but the **gender gap has been closing**
- Global science has become **more globalized** in terms of origins and destinations – but female researchers tend to **migrate shorter distances** and to a **narrower range of destinations**; but with persistent cross-national heterogeneity
- US was **top destination** for both men and women, but its dominance **has shrunk over time**, with **China emerging** as a prominent destination



Return migration of German-affiliated researchers: analyzing departure and return by gender, cohort, and discipline using Scopus bibliometric data 1996–2020

Xinyi Zhao^{1,2} · Samin Aref^{1,3} · Emilio Zagheni¹ · Guy Stecklov⁴

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© The Author(s) 2022

Abstract

The international migration of researchers is an important dimension of scientific mobility, and has been the subject of considerable policy debate. However, tracking the migration life courses of researchers is challenging due to data limitations. In this study, we use Scopus bibliometric data on eight million publications from 1.1 million researchers who have published at least once with an affiliation address from Germany in 1996–2020. We construct the partial life histories of published researchers in this period and explore both their out-migration and the subsequent return of a subset of this group: the returnees. Our analyses shed light on the career stages and gender disparities between researchers who remain in Germany, those who emigrate, and those who eventually return. We find that the return migration streams are even more gender imbalanced, which points to the need for additional efforts to encourage female researchers to come back to Germany. We document a slightly declining trend in return migration among more recent cohorts of researchers who left Germany, which, for most disciplines, was associated with a decrease in the German collaborative ties of these researchers. Moreover, we find that the gender disparities for the most gender imbalanced disciplines are unlikely to be mitigated by return migration

- A majority of disciplines are dominated by male researchers in Germany, however, international mobility may play in helping to moderate some of the most extreme gender disparities.
- International migration, especially returning after the experience of international migration can help prolong the academic life of researchers in Germany, however, male researchers still have a longer academic life than female researchers.
- Both the outward and the returnee subpopulations in most disciplines were more male-dominated, female migrant researchers are less likely to return to Germany relative to their male counterparts.



Global patterns of migration of scholars with economic development

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Edited by Ronald Lee, University of California, Berkeley, CA; received October 24, 2022; accepted December 13, 2022.

January 18, 2023 | 120 (4) e2217937120 | <https://doi.org/10.1073/pnas.2217937120>

We leverage metadata on over 36 million journal articles and reviews indexed by Scopus in order to estimate migration of scholars based on information on changes in their institutional affiliations over time. We produce a database of yearly international migration flows of scholars, for all countries from 1998 to 2017. We use the open-access database to provide descriptive evidence on the relationship between economic development and the emigration propensity of scholars. Statistical analysis using generalized additive mixed models reveals that emigration rates initially decrease as GDP per capita increases. Then, starting from around 25,000 dollars (2017 constant international dollars at purchasing power parity), the trend reverses and emigration propensity increases as countries get richer. This U-shaped pattern contrasts with what has been found in the literature for emigration rates for the general population and calls for theoretical frameworks to understand the heterogeneous responses of migration to development.

high-skilled migration | bibliometric data | economic development | migration of scholars

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This article contains supporting information online at <https://doi.org/10.1073/pnas.2217937120#supplementary-materials>.

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¹E.S., E.Z., M.J.D., T.T., and A.A. contributed equally to this work. The author order was determined randomly.

- The relationship between emigration and economic development has an inverse-U shaped pattern for the general population.
- For **scholars we observe the opposite pattern** (U-shaped).
- Emigration rate initially decreases with economic development, reaches an equilibrium and then increases.
- Richer countries offer relatively **higher number of available positions**, promote **hiring international researches** from.



Forthcoming in *Demography*

Homecoming after Brexit: evidence on academic migration from bibliometric data

Ebru Şanlıtürk

Samin Aref

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Francesco C. Billari

Abstract

This study assesses the initial effects of the 2016 Brexit referendum on the mobility of academic scholars to and from the United Kingdom (UK). We leverage bibliometric data from millions of Scopus publications to infer changes in the countries of residence of published researchers by the changes in their institutional affiliations over time. We focus on a selected sample of active researchers whose movements are traceable for every year between 2013 and 2019, and measure the changes in their international migration patterns. While we do not observe a brain drain following Brexit, we find evidence that the mobility patterns of scholars began to change following the referendum. Among the active researchers in our sample, we find that their probability of leaving the UK increased by approximately 86% if their academic origin (country of first publication) was an EU country. For scholars with a UK academic origin, we observe that after Brexit, their probability of leaving the UK decreased by approximately 14%, and their probability of moving (back) to the UK increased by around 65%. Our analysis points to a compositional change in the academic origins of the researchers entering and leaving the UK as one of the first impacts of Brexit on the UK and EU academic workforce.

- Following the Brexit referendum we observe a compositional change in the British academic environment.
- Scholars who **started their academic career in an EU country have a higher probability to leave the UK** after the Brexit referendum.
- Scholars who started their **academic career in the UK have a higher probability to return the UK** after the Brexit referendum.
- If this trend continues, British academia would **not risk brain drain** but might **become more insular**.

Scholarly Migration Database



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 - <https://twitter.com/SMDatabase>



Please tweet with hashtag #BiblioDemography,
a tribute to James W. Vaupel (1945-2022).
Thanks Ilya and Jonas for bringing up Jim's
labeling idea!
and please follow:
<https://twitter.com/SMDatabase>



THANK YOU!

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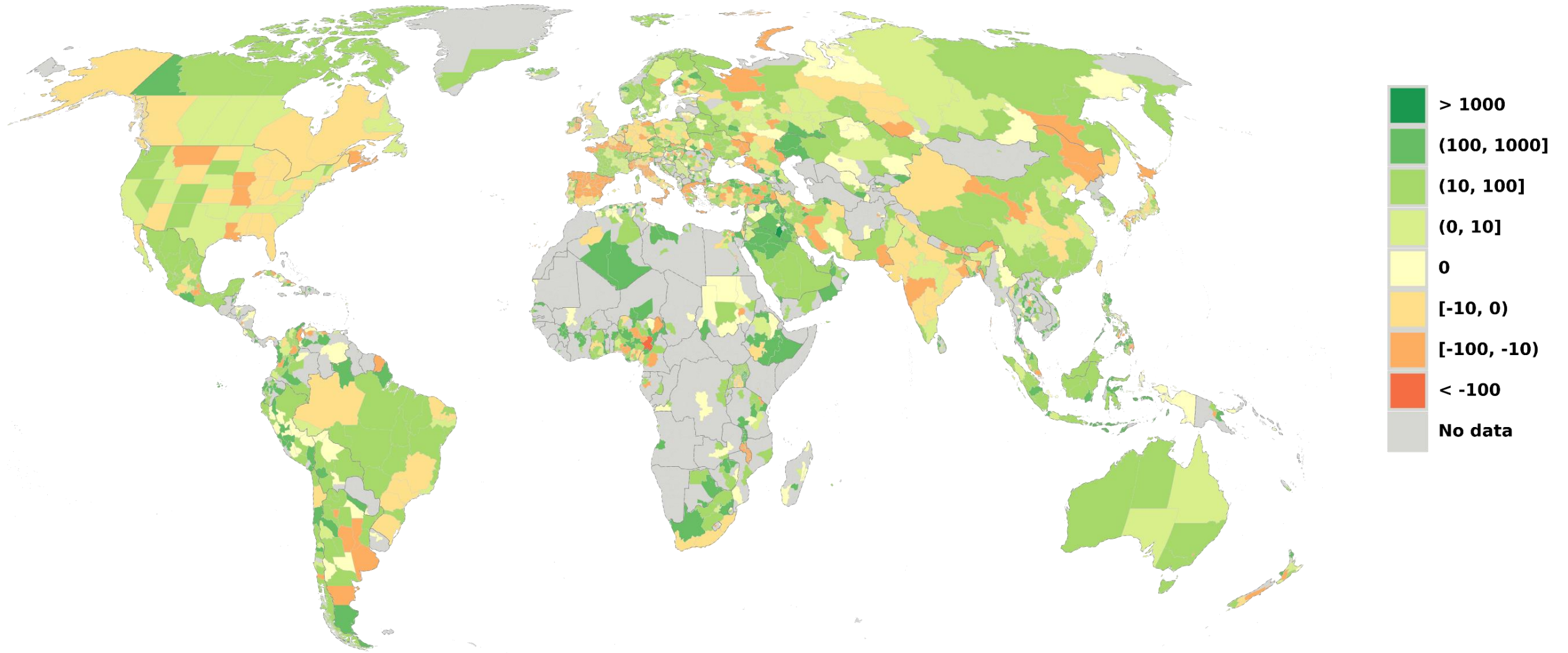
Executive Director

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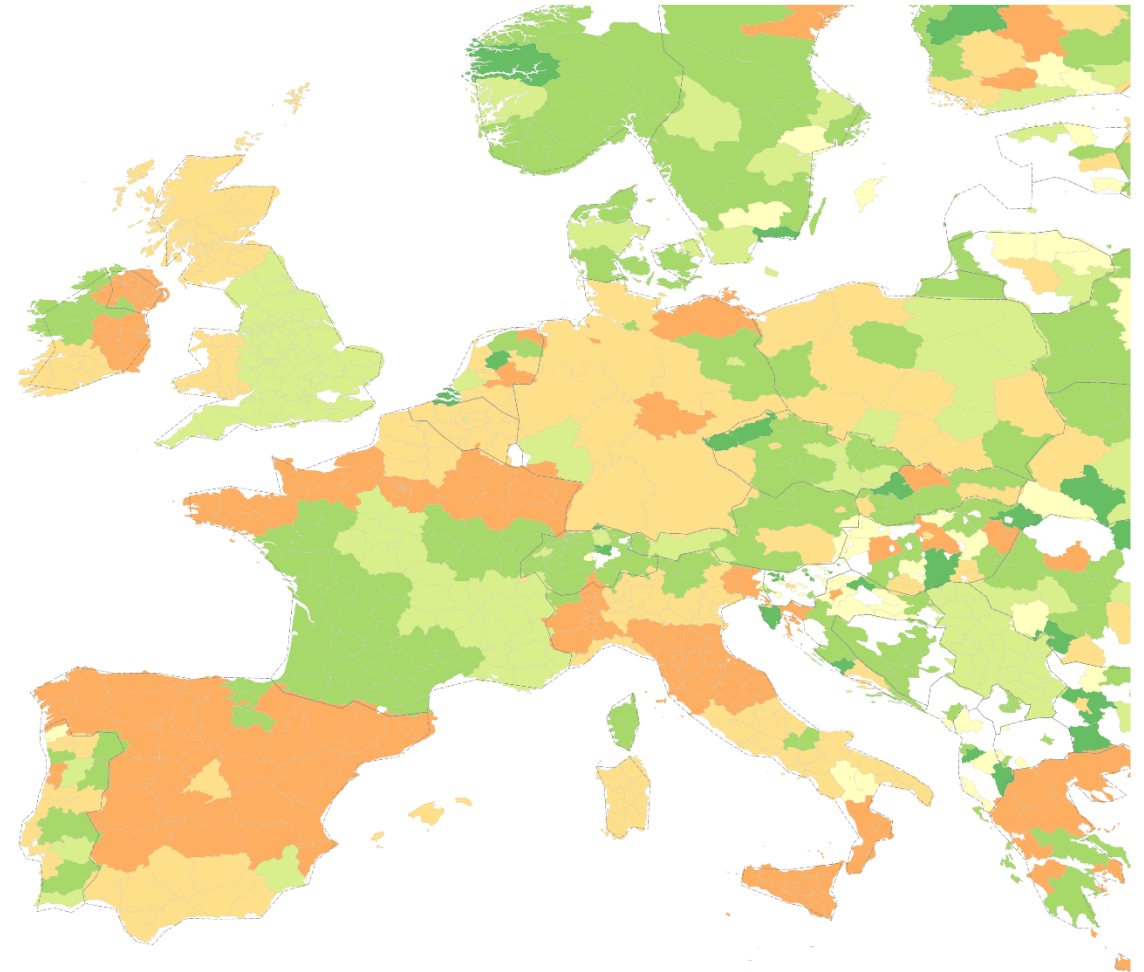
Additional slides, for Q&A





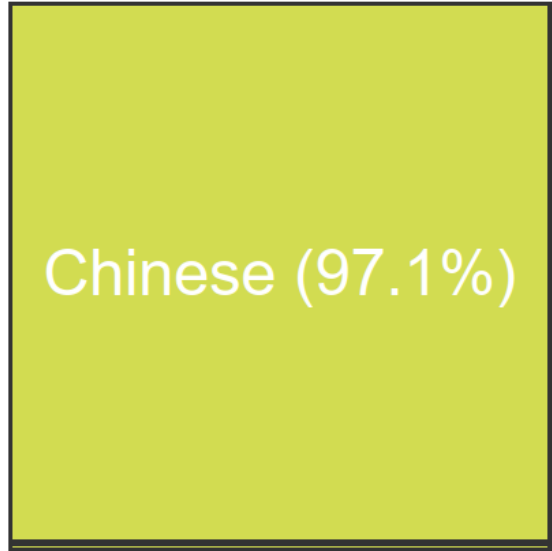
Combined internal and international net migration rate (NMR) per 1000 scholars at the sub-national level (GeoNames admin-1) worldwide (top) and Europe (right). Colors present the NMR of scholars sent (negative, red colors) or received (positive, green colors) in a year from 2012-2017.

REF: Akbaritabar et al. 2023

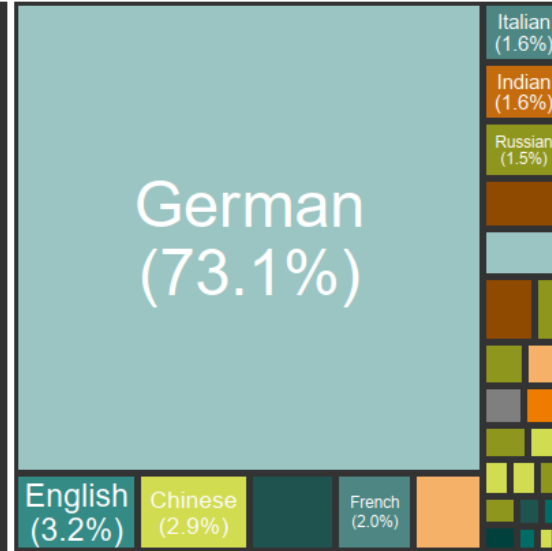




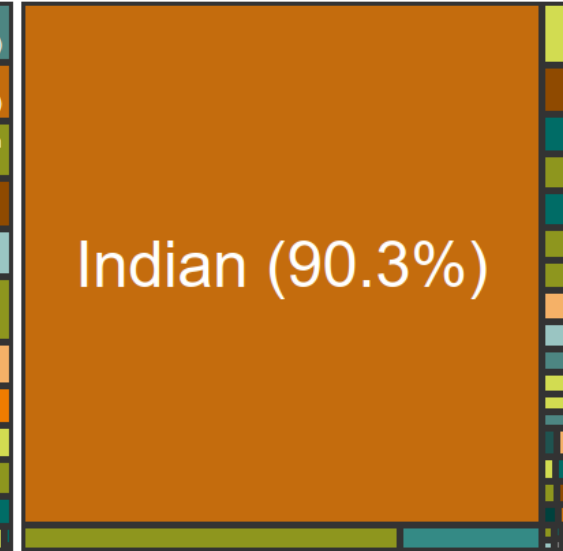
China



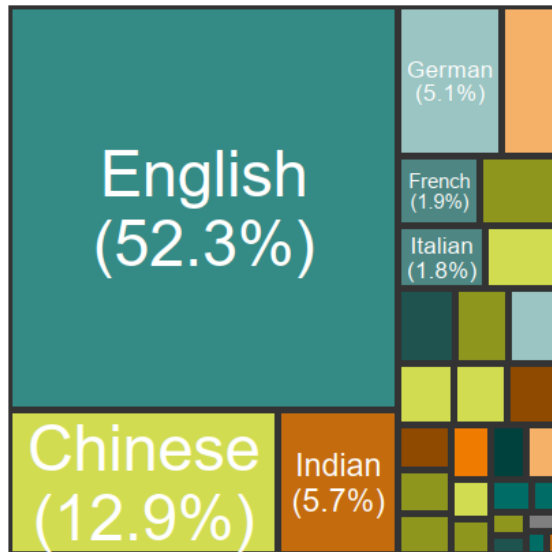
Germany



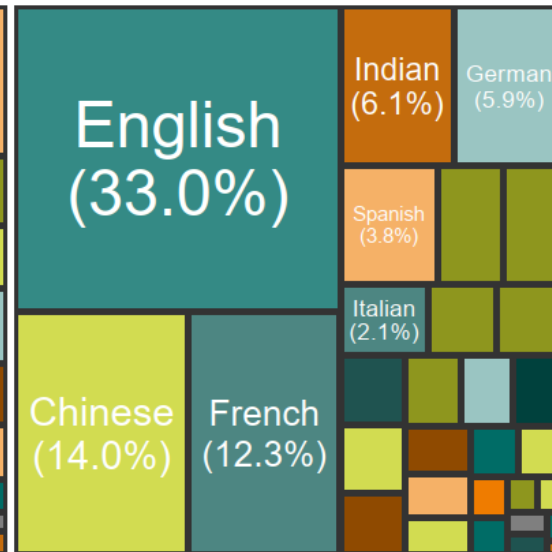
India



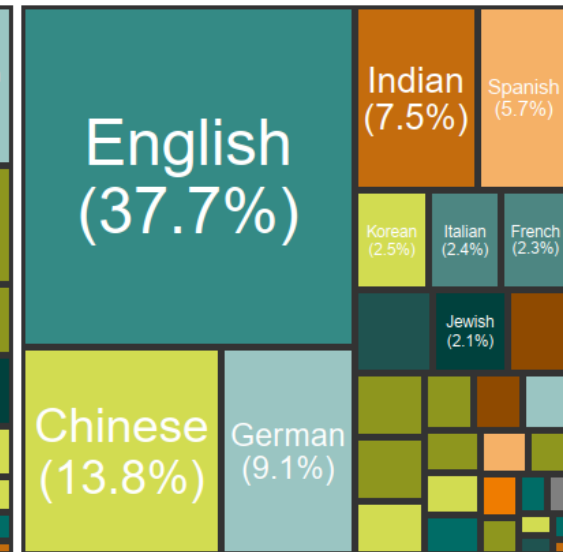
Australia

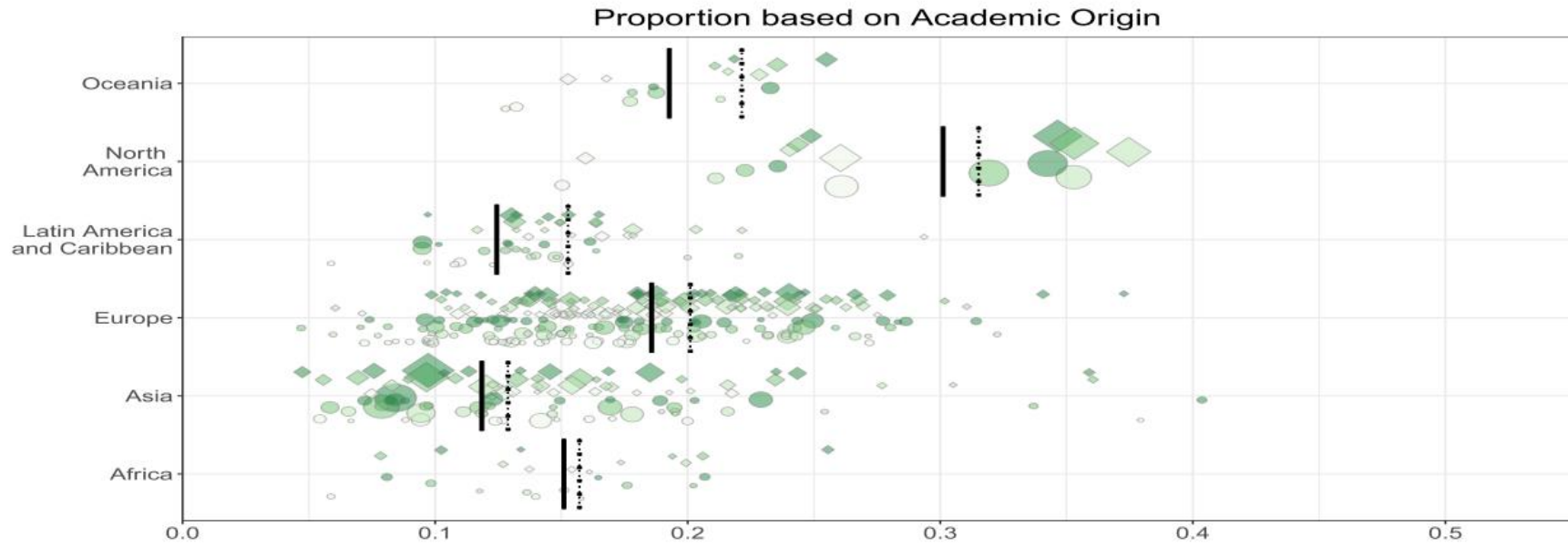
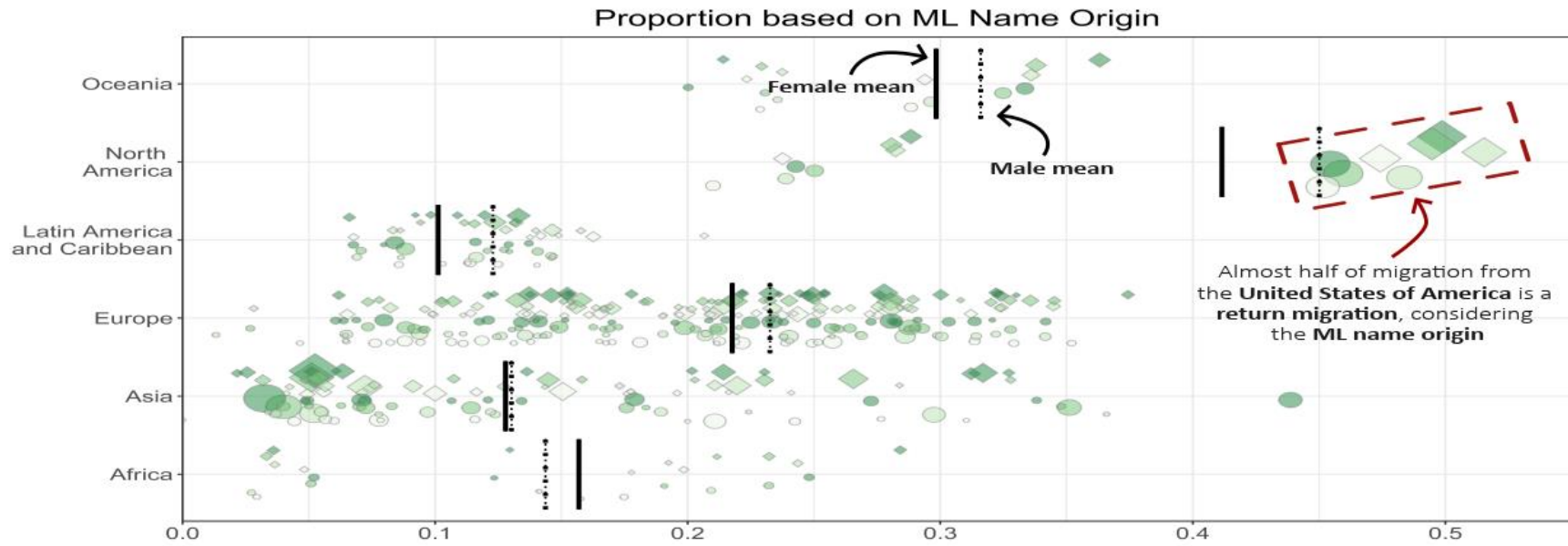


Canada



United States





GENDER DETECTION

